

PHYSICS

FORM 1

TERM	WEEK		MODULE	CHAPTER	LESSON TITLE	Nº OF PERIODS	
	1		MODULE 1 THE WORLD OF SCIENCE	Introduction to sciences + Scientific contributions and discoveries	First contact with the learners, definition and branches of science.	1	
	2				Prominent scientists and discoveries and contributions to improvement in the lives of humans.	1	
				Introduction to Physics	Definition of physics and some of its branches. What physicists do and how. Observing nature and seeking comprehension.	3	
	3			Basic laboratory equipment	Some basic equipment in the Physics laboratory	1	
				Safety laboratory rules	Safety rules for working in the Physics laboratory	2	
					Job opportunities for science students		
	4			Introduction to measurement.	Simple measurements using measuring instruments/ INTERGRATION LESSON	2	
	5			EVALUATION			2
	6			Physical quantities	Identifying physical and non-physical quantities	2	
	7				Some Units of measurement and S.I units		
				MODULE II MATTER: PROPERTIES AND TRANSFORMATION	matter	-States of matter and differences between them	2
	8		States of matter		-interconversion processes	2	
	9		Measurement of length		– Define length & state its S.I and sub units Measurement of length	2	
	10		Measurement of mass		– Define mass and state its SI and sub units Measurement of the mass of a body /INTERGRATION LESSON	2	
	11		EVALUATION			2	

	12		Measurement of weight	Define weight and state its units Differentiate between mass & weight Measurement of weight	2
	CHRISTMASS BREAK				
	13		Measurement of volume	Define volume & state its units -measurement of volumes of liquids, regular & irregular solids	3

14			Measurement of Density	- Define density as the mass per unit volume and state its units	2	
15			Measurement of Temperature	-Define temperature and state its S.I and sub units/conversion	1	
			Using information on products	-safety rules on products/ materials	1	
				-using information		
16		MODULE III ENERGY : APPLICATIONS AND USES	Forms of energy	-Definition, forms, sources of energy	1	
17			Energy needs	-Daily applications of energy	2	
18				-common devices that use different forms of energy and principle of energy conservation. INTERGRATION LESSON		
			EVALUATION			2
19			Solar energy	- Components and uses of solar energy	1	
			Chemical energy	-sources and uses of chemical energy	1	
20			Electrical energy	-sources and uses of electrical energy	1	
			Thermal energy	-sources and uses of heat.	1	
21				-conduction (conductors / insulators) -convection / application (ventilation) -radiation (good and bad absorbers/emitters)	2	
22			Forces and motion	Definition and effects of forces	1	
				-definition/types of motion	2	
23				-safety rules: uses of seat belts, low speed/road signs	1	
24				INTERGRATION LESSON + EVALUATION	2	
EASTER HOLIDAY						
25		MODULE IV HEALTH EDUCATION	Sound	Definition and production	1	
26			Sound	The ear and sound perception	2	
27			Detection of sound	-Effects of loud sound and prevention	1	
28			Temperature	-measurement of temperature of a body (clinical thermometer) -normal and abnormal body temperatures	2	
29			Sports and physics	Body posture: importance of good posture to the body	2	
30		MODULE V ENVIRONMENTAL EDUCATION AND SUSTAINABLE DEVELOPMENT	-RADIATIONS	-Harmful waste & background radiation -handling radioactive substances/ INTERGRATION LESSON	2	
31			EVALUATION			2
32			-global warming And climate change	-greenhouse effect (concept/causes) -climate change -environmental sustainability	2	
33			Common tools	Machines (identification and uses)	2	

	34		MODULE VI	Maintenance	-lubrication, cleaning and repairs using the tools	2
	35		TECHNOLOGY	Technical drawing	Instruments used/ sample drawings INTERGRATION LESSON	2
	36			EVALUATION		2

NB: This National Harmonized Progression could be modified in terms of; lesson titles, time allocated for lessons and order of succession of lessons. Such modifications need to be pointed out and discussed with Regional and National Pedagogic Inspectors so that the modifications are communicated to go into effect nationwide as from the same time.

PHYSICS

SUBJECT: PHYSICS

FORM 2

Term	Week		Module	Chapters	Lesson title	Duration	
I	1		Module 1:The world of science	First contact with students and presentation of scheme of work		1H	
	2			1: Scientific method part 2	1.0 Collecting data-importance of data	2H	
	3			1: Scientific method part 2	1.2 Interpreting data and concluding	2H	
	4			2:Simple application of measurements	1.3 Predicting and evaluating 1.4 Planning (recall basic quantities units and instruments)	2H	
	5			2:Simple application of measurements	1.5 Measurements of speed and units 1.6 Measurement of density and Units	3H	
	6			Revision and integration - FIRST EVALUATION			2H
	7		Module 2 : matter, Properties and transformation	3 : Change of state	2.1 Physical state of matter 2.2 Characteristics of matter in the different states.	2H	
	8			4 : Temperature	2.3 Temperature – measurement and units and effects on matter	2H	
	9			5:Thermal and electrical insulation	2.4 Thermal and electrical insulation	2H	
	10			6:Action of heat on materials	2.5 The use of a candle	2H	
	11			7:Action of heat on materials	2.5 Cooling	2H	
	12			INTEGRATION AND EVALUATION			2H
			CHRISTMASS BREAK				
II	13		Module 2 : matter, properties and transformation	8: Action of electrical energy on materials	2.6. The use of a bulb in a simple electric circuit.	2H	
	14		Module 3: energy, value and uses	9: Energy needs of human beings	3 Sources and uses of energy. transmission of energy	2H	
	15			10 :Renewable energy	-Solar panel for heating -Other sources	2H	
	16			11:Electricity	Electricity for the home and the simple electric circuit	2H	
	17			12:Light	Sources of Light	2H	
	18			INTEGRATION AND EVALUATION			
	19			12: Light	- Types of light receivers		2H
	20				-Beams and shadows		2H
	21			13:Energy exchange	Linking one form of energy to other forms		2H
	22		14: Motion	Distance, time and speed		2H	
	23		Module 4 : Health education	15: Distribution of pressure in a liquid	4. Average blood pressure	2H	
	24			INTEGRATION AND EVALUATION			2H
			Easter holidays				
III	25		Module 4 : Health education	16: muscle stress	Sports and physical education	2H	
	26+ 27			17: Types of lenses and their uses.	The eye as an imaging device and use of lenses to aid eyes with vision defects	3H	
	27		Module 5: Environmental	18:Radiation	5.0 Introduction 5.1 Radiation emitted into the atmosphere -Cosmic waves from the sun	2H	

	28		Protection and sustainable development	19: Weather and communication	5.2 Global warming and Climate change	2H	
	29			20: The Greenhouse Effect	5.2 Global warming and Climate change	2H	
	30			INTEGRATION AND EVALUATION			2H
	31						
	32		Module 6: Technology	22:introduction to technology	6.0: Introduction to technology Review of form one	2H	
	33			23 : Project	6.1 Fabrication of common instruments	2H	
	34			24 Care and maintenance	6.2 Care and maintenance Principle of functionality of some common appliances	2H	
	35			25:Technical drawing	6.3 Technical drawing	3H	
	36			GENERAL REVISION AND EVALUATION			

NB: This National Harmonized Progression could be modified in terms of; lesson titles, time allocated for lessons and order of succession of lessons. Such modifications need to be pointed out and discussed with Regional and National Pedagogic Inspectors so that the modifications are communicated to go into effect nationwide as from the same time.

TERM		WEEK		MODULE	CHAPTER	LESSON TITLE	No of periods
FIRST TERM		1		MODULE 1: INTRODUCTION TO MECHANICS	1. Physical quantities	- First contact with the students; Definitions/examples / units of physical quantities. - Scalar physical quantities and vector quantities	02
						- Prefixes and use of standard form - Name some basic equipment used in the study of force: newton meter, balance, springs, masses etc.	01
						- Safety rules for working with different equipment - comparison between mass and weight	02
		2				Measurement of mass/Conversions between different units of mass	01
						Measurement of volume, temperature and time	02
		3		MODULE II: MATTER: PROPERTIES AND TRANSFORMATION	2. Density	Definition / calculation / unit of density	01
						Describe and carry out experiments to measure the density of regular & irregular objects	02
						Applications of the density of a material in engineering works	01
		4			3. Pressure	- Definition / calculation / unit of pressure. - Factors that affect pressure in solids - Application of pressure in solids	02
						- Pressure in liquids - Calculation / Factors affecting of pressure in liquids - Characteristics of pressure in liquids - Applications; Hydraulic machines	02
						INTERGRATION / EVALUATION / REMEDIATION	02
		5					
		6				Atmospheric pressure and weather	02

						– Demonstration/ Measurement/Application of atmospheric pressure – Pressure and health (Effects of high/low pressure)	
						- Effects of pressure on boiling point. -Application of high pressure	01
		7			4. Elasticity and Hooke’s law	– Definition of elasticity – Elastic and non-elastic materials – Hooke’s Law and elastic limit	02
						– Sketch F – e graphs within the elastic limit. – Experimental demonstration of Hooke’s law – Describe situations in which Hooke’s law applies	02
		8		MODULE III: ENERGY: APPLICAT IONS AND USES	5. Forms of energy	– Definition / Forms of energy	01
						– Sources of energy. – Renewable and non-renewable energy sources	01
						– The law of conservation of energy / Application – Transducers and energy flow diagrams.	02
		9 +1 0			5. Forms of energy	– Energy transformations – Calculations of potential (gravitational and elastic) and kinetic energies	02
	6. Work			– Definition / Calculation of work involving force and displacement (in the same direction only) – Examples of situations where work is done		02	
				7. Power	– Definition / calculation of power – Power ratings of some devices e.g. electric iron, light bulbs, electric motors etc.	01	
				12		INTERGRATION / HARMONISED EVALUATION / REMEDIATION	
					8. Simple Machines	– Definition / advantages of a machine.	01
						- MA / VR / Efficiency. Relationship b/w the 3	02
			13		8. Simple Machines	– Simple machines (Lever, inclined plane and pulley, hydraulic machines) / Experiments and Calculations	02
S				MODULE IV:	9. Reflection of light	– Description / Propagation of light – Rays and types of beams	02

				OPTICS		– Luminous and non-luminous sources of light – Transparent, translucent and Opaque objects – Shadows and applications	
		15				– Definition / laws of reflection of light – Experiment to show that $i = r$	01
						– Image formation by plane mirrors. – Characteristics of images formed by plane mirrors – Uses / Simple calculations on plane mirrors – Curved mirrors(Types and applications)	02
					10. Refraction of light	– Identification of refraction phenomena in everyday life / Definition of refraction	01
						– Laws of refraction	01
						– Refractive index and speed of light	01
					11. LENSES	– Experiment to determine refractive index of a glass block	01
		16				– Effects of refraction of light in everyday life – Real/apparent depth and refractive index	01
						– Total internal reflection (TIR) – Conditions / Daily occurrence / Definition of c – Ray diagram to show TIR	02
						– Use of TIR / Solve problems using $\text{sinc} = \frac{1}{n}$	01
		17				INTERGRATION / EVALUATION / REMEDIATION	03
		18				– Lens action/Types of lenses/Definition of terms:	01
		19				– Ray diagrams to illustrate the meaning of principal foci, for converging & diverging lenses – Measurement of focal length of a converging lens	03
						– Image formation by converging/diverging lenses – The lens formula and magnification, $m = \frac{v}{u}$	02
		20					
		21		MODULE V:		– Definition / Demonstration of dispersion.	01

THIRD TERM				EM SPECTRUM	12. Dispersion of light.	– Formation of a pure spectrum from white light. – Natural occurrence of the dispersion	01 01	
		22			EM Spectrum	INTERGRATION / EVALUATION / REMEDIATION		03
		23				– Relative positions of radiation on EM spectrum, in terms of wavelength and frequency	01	
						– Methods of detection/properties radiations – Health hazards caused by EM waves	02	
		24		MODULE VI: Projects and Elementary Engineering	13. Understanding how to go about a Project.	– Technical drawing: General notions involved – 2-dimensional diagrams/orthogonal representation	03	
		BREAK				EASTER BREAK		
	– Technical Project Types /Elements involved – Definition/Steps involved in planning a project					03		
		25				– Study the advantages offered by the project; – Feasibility Studies – Theoretical knowledge of a simple technical object	03	
		26				Different energy components when using the instrument;		
		27				– Technical diagram / studies – Identification of the movement of certain parts with respect to others (guide, reducing friction)	03	
		28			14. Investigating forces	– Definition of new words – Realisation of a project. -Understand the uses of measuring instruments and their limitations. Understand why and how things move. – Understand the relationship between mass and volume through measurements	03	
		29			INTERGRATION / EVALUATION / REMEDIATION			
		30				– Observing colours of light on a CD or DVD and suggesting reasons for their appearance. – Observe a pen or pencil in a glass of water and explain why there is an image seen with the object.	03	

		31				– Use the image to estimate the refractive index of water.	03
		32 - 35				GENERAL REVISION / END OF YEAR EXAMINATION	12

Term	Week		Module	Chapters	Lesson title	Duration
I	1		MODULE 1 ENERGY: APPLICATION AND USES	First contact with students and presentation of scheme of work		2H
				Heat	1.1 concept of heat and temperature 1.2 measurement of temperature	
	2			Thermometry	1.3 Thermometry 1.4 liquid-in-glass thermometer 1.5 calibration of temperature using fixed points 1.6 clinical and normal laboratory thermometers	3H
	3			Calorimetry	1.1 Calorimetry 1.2 Heat capacity and specific heat capacity Calculation involving $Q=mc\Delta\theta$	4H
	4			Latent Heat	1.3 Latent Heat and specific Latent heat 1.4 Cooling effect	3H
	5			Heat transfer	1.5 Heat Transfer -Conduction, Convection and Radiation	3H
	6	INTEGRATION AND EVALUATION		3H		
		Expansion			1.6 Thermal Expansion -The bimetallic strip and its principles -Radiant energy converters	
	7		Properties of Waves	2.1 Definition and classification of waves 2.2 Properties of waves: Reflection, Refraction, Diffraction and Interference 2.3 Calculations using the equation $v=f\lambda$	4H	
	8		Stationary Waves	2.4 Stationary waves 2.5 Harmonics and Overtones 2.6 Relationship between inter-node distance and wavelength	3H	
	9		MODULE 2 : WAVES	Sound Waves	2.7 Production and transmission of sound 2.8 Characteristics of sound (Amplitude, frequency, and pitch, quality and overtones. Frequency limit of audibility) 2.9 Measurement of the speed of sound (echo method and gun firing method) Application of echo in sea bed 2.10 Calculation of speed of sound in air	2H
	10				2.9 Measurement of the speed of sound (echo method and gun firing method) Application of echo in sea bed 2.10 Calculation of speed of sound in air	4H
11		Vibration in strings		2.11 Vibrating strings Relationship between frequency and length, frequency and mass per unit length, frequency and tension	2H	
12		Forced vibration		INTEGRATION AND EVALUATION		3H
			Forced vibration on a string and in a tube			
			CHRISTMASS BREAK			
II	13		MODULE 2 : WAVES	Resonance	Define and explain resonance. -State importance and applications of resonance including measurement of speed of sound	4H
	14		MODULE 3 ELECTRICAL ENERGY	ELECTROSTATICS	Charges and their origins	1H
	15			The coulomb	3.1 Types of charge and testing for charge with electroscope 3.2 Charging and discharging 3.3 Separation of charges	2H
					3.4 conductors and insulators 3.5 Basic law of electrostatics and Coulomb's law	3H
16		3.6 Force between charges Positive and negative charges Application of electrostatics industrially eg photocopying, painting		3H		
17		Current Electricity	3.8 Introduction 3.9 Electricity and charge movement. Use of meters in electric circuits, how they are connected and reasons..	4H		
18		INTEGRATION AND EVALUATION				3H

	19			emf	3.10 Emf and pd 3 sources of emf	3H
	20				3.11 Energy consumption $W=QV$ and $P=VI$	3H
	21				3.12 ELECTRIC CIRCUITS -components of a circuit network	3H
	22				3.13 Ohms' law and Resistance	2H
	23				3.14 Circuit network -Parallel and series connections Calculations in circuits	3H
	24				INTEGRATION AND EVALUATION	3H
				EASTER HOLIDAYS		

Term	Week		Module	Chapters	Lesson title	Duration
III	26		MODULE III ELECTRICAL ENERGY	Power in ac and dc	3.15 DC and AC Definition Transportation of electric current	3H
	27				3.16 calculation of power dissipated	3H
	28			House wiring	3.17 Calculation of power consumed at home (KWH) 3.18 Fuse and selection	3H
	29				3.19 House wiring 3.20 Ring circuit and linear circuit in house wiring 3.21 Safety precaution	3H
	30		MODULE IV PROJECTS AND ELEMENTAY ENGINEERING	CRO	3.22 The cathode ray oscilloscope	3H
	31			Technical Drawing	INTERGRATION AND EVALUATION.	3H
	32				Realization of a technical project. 1 Technical drawing 2 Reading of technical drawing	3H
	33			Section view Building plans	1.2. Cross section of an Object	
	34				List the various ways of viewing an object. e.g top – view	3H
	35				Reading of the plan of a construction sheet Drawing and giving of dimensions of a building	3H
	36				INTERGRATION AND EVALUATION.	3H
					GENERAL REVISION	3H

NB: This National Harmonized Progression could be modified in terms of; lesson titles, time allocated for lessons and order of succession of lessons. Such modifications need to be pointed out and discussed with Regional and National Pedagogic Inspectors so that the modifications are communicated to go into effect nationwide as from the same time.

Term	Week		Module	Chapters	Lesson title	Duration
I	1		Module 1 : FIELDS: MAGNETIC FIELDS AND THEIR EFFECTS	First contact with students and presentation of scheme of work		1H
	2			Magnetic Field	1.0: Magnets 1.1 Introduction to magnetism- magnetic and non-magnetic substances -Identification of magnets 1.2 Law of magnetism - applications of magnets - Making magnets and care for magnets -Hard and soft magnetic materials	4H
	3				1.3 Magnetic Field 1.4 Magnetic flux pattern -defining magnetic flux -Drawing of magnetic field lines for; bar magnet, the earth and the horse-shoe magnet. -flux pattern around two poles of magnets near each other	3H
	4			Magnetic effect of current	1.5 Magnetic effect of current -Magnetic field pattern of a straight conductor carrying current -direction of field lines: current carrying solenoid and factors which affect field strength. Force on a current carrying conductor placed in a magnetic field -factors which determine the size of the force. Principle of electric motor	4H
	5			Revision and integration		3H
	6			FIRST EVALUATION		3H
	7, 8			Electromagnetic Induction and Alternating Current	1.6 Electromagnetic Induction -introduction to electromagnetic induction - Faraday's law -Lenz laws -Conservation of energy	5H
	9, 10				1.7 Alternating Current Mutual inductance The transformer -Energy losses and remedies -Turn ratio. -Efficiency -Practical transformers and power transmission -Applications of transformers	6H
				Revision and integration		3H
	12			HARMONIZED EVALUATION		3H
	13		MODULE 2 ENVIRONMENTAL PROTECTION:	The Atom	2.0 The nuclear model of the atom - the composition of the atom The electron (Q=Ne)	3H

	1 4		Modern Physics and basic electronics.		2.1 The nuclear model of the atom - the composition of the atom The electron (Q=Ne)	2H	
			CHRISTMASS BREAK				
II			ENVIRONMENTA L PROTECTION: Modern Physics and basic electronics.				
				The Nucleus	2.2 The Nucleus Nuclear stability 2.3 Radioactivity and decay equations	4H	
	1 6			Radioactivity	2.4 Radioactive decay α , β and γ . And their properties including behaviour in electric field, magnetic field and cloud chamber. 2.5 The concept of half-life -The importance and use of isotopes. - Background radiation	4H	
	1 8			Revision and integration + EVALUATION		3H	
	1 9				2.6 uses of radioactive isotopes: eg in medicine, agriculture. 2. 7: safety and hazard of radioactivity	3H	
				Basic electronics	2.8 Semiconductors -intrinsic and extrinsic -P-type and N-type P-n junctions and rectification	3H	
					INTEGRATION	3H	
	2 0			EVALUATION		3H	
	2 1		Module 3: MECHANICS	Physical quantities	3.1 Vector and scalar physical quantities 3.2 Forces -Type of forces -Resolution of forces -Free body diagrams -Archimedes Principle	3H	
	2 2				3.3 Mass and weight Turning effect of forces -Moments, couples and applications	3H	
	2 3			Motion	3.4 Linear motion - Distance, displacement, Speed, velocity, acceleration -motion graphs 3.5 Uniform motion -Equations of uniformly accelerated linear motion and simple calculations - Free fall and gravity -Experiment to determine g, by free fall	4H	
	2 4			Newton's law	3.6 : linear momentum -Interaction and conservation of momentum. -Principle of conservation of momentum and real life situations 3.7 Newton's laws of motion	4H	
				EASTER HOLIDAYS			

Term	Week	Module	Chapters	Lesson title	Duration
III	25	Module 3: MECHANICS	INTEGRATION		3H
	26		EVALUATION		3H
	27	Module 4: PROJECTS AND ELEMENTARY ENGINEERING		1 Preservation and Maintenance of appliances.	2H
	28		Preservation of appliances	The essential elements in a repair box. (give the name, role and method of application)	2H
	29			INTEGRATION	3H
	30		Maintenance of appliances	Understand the labelling on appliances	3H
	31			Techniques of dismantling and assembling of appliances	3H
	32			INTEGRATION	3H
	33			Techniques of dismantling and assembling of appliances (FOLI AND LIFO) techniques	2H
	34			INTEGRATION	2H
	35			EVALUATION	
	36		GENERAL REVISION		

NB: This National Harmonized Progression could be modified in terms of; lesson titles, time allocated for lessons and order of succession of lessons. Such modifications need to be pointed out and discussed with Regional and National Pedagogic Inspectors so that the modifications are communicated to go into effect nationwide as from the same time.

SUBJECT: PHYSICS
LOWER SIXTH SCIENCE

Term	Week	Module	Chapter	Lesson title	Duration	Module	Chapter	Lesson title	Duration
I	1	Module 1: Physical quantities		First contact with students and presentation of scheme of work/Advanced Level course outline. Physical quantities	02H	Module 3: Energetics		First contact with students and presentation of scheme of work/Advanced Level course outline	02H
			1.1 Physical quantities	Physical quantities, base and derived physical quantities/units, SI units	02H				
	2			Dimensions, physical equations and homogeneity of physical equations.	04H		3.1 Temperature	Thermal equilibrium, zeroth law and thermometric properties	02H
	3			Experimental physics. Experimental techniques, approach, accuracy, sensitivity error and precautions	04H			Temperature measurement, temperature scales and disagreement between temperature scales	02H
	4		2.1 Kinematics	Motion, distance/displacement, speed/velocity, Linear momentum, acceleration	04H			Different types of thermometers	02H
	5			Graphs of motion in one dimension and equations of uniformly accelerated motion	04H			Different types of thermometers	02H
	6			INTEGRATION EXERCISES AND EVALUATION					
	7			Motion under gravity and experiment to measure acceleration due to gravity	04H		3.2 Energy Transfer	Heat capacity and specific heat capacity	02H
	8			Projectile motion **This could still be taught under motion in the gravitational field.	04 H			Experiments to measure specific heat capacity	02H
	9		2.2 Dynamics and force	Meaning and nature of forces, types and classification. Centre of gravity, centre of mass, free-body diagrams and resultant of coplanar forces	04H			Latent heat and specific latent heat	02H
	10			Turning effect of forces, moment and couples, integration exercise	04H			Experiments to measure specific latent heat of fusion and specific latent heat of vaporization	02H
	11			Equilibrium of objects acted upon by a number of coplanar forces	04H			Heating and cooling curves, integration exercises	02H
	12	HARMONISED EVALUATION							

II	13			Newton's first and second Laws, impulse, experimental investigations of $a \propto F$ and $a \propto \frac{1}{m}$	04H		3.2 Energy Transfer	Thermal conduction and mechanism, temperature gradient	02H
	14		2.3 Newton's Laws of motion and momentum	Newton's Third law of motion, the law of conservation of linear momentum from Newton's second and third laws	04H			Thermal conductivity,	02H
	15			Law of conservation and experiment to investigate the law of conservation of linear momentum.	04H			Comparing electrical conduction to thermal conduction, experiment to compare thermal conductivity of different materials	02H
	16			Elastic and inelastic collisions.	04H			Thermal convection and radiation, mechanisms	02H
	17			Explosions, head-on and oblique collisions.	04H		3.3 Electrical energy	Electric current, electric potential difference, drift velocity.	02H
	18			INTEGRATION EXERCISES AND EVALUATION					
	19		2.4 Work, energy and power	Work, potential energy and kinetic, Law of conservation of mechanical energy	04H		3.3 Electrical energy	Electrical resistance, resistor colour codes, Ohm's Law and resistivity.	02H
	20			Work-kinetic energy theorem, efficiency, power	04H			Experiment to determine resistivity, temperature coefficient of resistance	02H
	21		2.5 Circular motion	Period and frequency, angular speed and velocity, centripetal acceleration and centripetal force, consolidation examples	04H			Resistor networks and the potential divider	02H
	22			Motion in a vertical circle, the conical pendulum, cornering and banking	04H			Ideal and non-ideal ammeters and voltmeters in circuit	02H
	23		2.6 Simple harmonic motion	Meaning and equations	04H		3.3 Electrical energy	Electromotive force, terminal p.d. and internal resistance	02H
	24			HARMONISED EVALUATION					
III	25		2.6 Simple harmonic motion	Energy changes in simple harmonic motion, examples of simple harmonic oscillators	04H		3.3 Electrical energy	Kirchhoff's Laws	02H
	26			Mechanical oscillations and resonance	04H			Potentiometer	02H
	27	Module 4: Matter, effects of energy	4.1 Solids and liquids	Differences in the molecular properties of solids, liquids and gases, molecular spacing, intermolecular force vs separation curves, potential energy vs separation curves	04H			Wheatstone bridge circuit, consolidation exercise	02H

	28	and applications		Elasticity and Young modulus energy stored in a stretched wire	04H	Option 1: Energy Resources and environmental Physics	3.4 Energy resources and environmental Physics	Classification of energy sources, functional energy, efficiency of conversion	02H
	29			Experiment to determine Young Modulus, consolidation exercises	04H			Hydroelectricity and wind energy	02H
	30			Surface tension and capillarity, Angle of contact, experiment to measure surface tension of water	04H			Solar energy and tidal energy	02H
	31		4.2 Gases and thermodynamics	Brownian motion in gases, gas laws and the ideal gas equation	04H			Biomass, geothermal energy and wave energy	02H
	32		INTEGRATION EXERCISES AND EVALUATION						
	33		4.2 Gases and thermodynamics	Kinetic theory of ideal gases, derivation of $P = \frac{1}{3} \rho \overline{c^2}$ and k.e. = $\frac{1}{2} m \overline{c^2} = \frac{3}{2} kT$.Distribution of molecular speeds, Real gases and Andrew's experiment	04H		3.4 Energy resources and environmental Physics	Fossil fuels and nuclear fuel.	02H
	34			First law of thermodynamics; isochoric, isobaric, isothermal and adiabatic processes	04H			Simple structure of the atmosphere, ozonosphere and the ionosphere, destruction of the ozone layer	02H
	35			Second law of thermodynamics, basic function of heat engines, entropy, Consolidation examples	04H			Greenhouse effect, global warming and climate change	02H
	36		END OF YEAR HARMONISED EVALUATION						

NB: This National Harmonized Progression could be modified in terms of; lesson titles, time allocated for lessons and order of succession of lessons. Such modifications need to be pointed out and discussed with Regional and National Pedagogic Inspectors so that the modifications are communicated to go into effect nationwide as from the same time.

SUBJECT: PHYSICS

UPPER SIXTH SCIENCE

Term	Week	Module	Chapter	Lesson title	Duration	Module	Chapter	Lesson title	Duration
I	1	Module 5: Field Phenomena	First contact with students and presentation of scheme of work/USS course outline		02H	Module 6: Waves around us.	First contact with students and presentation of scheme of work/USS course outline.		02H
			Newton's law of universal gravitation and inverse square relation				6.1 6.1 MECHANICAL WAVES: Classification of Waves: mode of propagation and medium of propagation with specific examples.		
	2		5.1 GRAVITATIONAL FIELDS	- Kepler's laws, Qualitative description of the earth's gravitational field. - Field Strength	04H		6.1 MECHANICAL WAVES	- The progressive wave and equation. - Graphical representation of waves.	02H
	3			- Variation of g inside and outside the earth. - Gravitational potential and gravitational potential energy	04H		6.1 MECHANICAL WAVES	Properties of Waves Production of waves, reflection, refraction, diffraction, Interference. Single slit pattern	02H
	4			- Escape velocity - Orbital speed and geostationary satellites. Orbital speed and movement of satellites with geostationary satellites as example.	04H		6.1 MECHANICAL WAVES	Double slits and multiple slits interference patterns and measurement of wavelength of a wave.	02H
	5		ELECTROSTATICS + ELECTRIC FIELDS	. Motion in the gravitational field: Projectile motion. **Check if this was not treated under mechanics in LSS.	04H			Properties of Waves: Polarization- meaning and production of plane polarized waves.	02H
				- Electric charge and current - Good and bad conductors of electricity. - Charging by friction, contact, induction and by chemical action. - Point action and the lightening conductor	02H + 02 H		INTEGRATION EXERCISES AND EVALUATION		
	6						3.2	The factors affecting the speed of transverse waves on taut strings and wires	02 H
	7			- Coulomb's law and inverse square nature. Dependence of size of force on medium. - Electric field and field strength	04H			Doppler Effect for Sound in air with specific cases of moving source/ stationary observer and moving observer with stationary source.	04 H
	8			- Electric potential and work done in moving a charge in an electric field. - Electric field dipole and torque.	04 H			Stationary waves and characteristics.	02H
	9		CAPACITORS	- Identification of capacitors and circuit symbols. - Measurement of capacitance. - Factors which affect capacitance.	04H				

				Relative permittivity					
	10			- Combination of capacitors (series & parallel) - Energy/charge stored in a different combinations of capacitors connected to a dc power supply	04H			Measurement of the speed of sound in air.	02H
	11			Charging and discharging capacitors through resistors and time constant. The equations for charging/ discharging and interpretations at t=0 and as $t \rightarrow \infty$.	04H			- Electromagnetic waves and their characteristics. - EM-spectrum; production, detection and uses of different sections	02H
	12			HARMONISED EVALUATION					
II	13		5.4 MAGNETIC FIELDS	- The magnetic field - Magnetic flux density and its units (the tesla). - Field patterns of current-carrying conductors (straight wire, plane circular coil, solenoid) - Force on a current-carrying conductor in a uniform magnetic field. - The direction of the force from Fleming's left hand rule	04H		3.2	- X-rays, production and uses - Meaning and application of plane polarized em waves.	02H
	14		5.4 MAGNETIC FIELDS	- Forces on objects in cross-fields - Torque on a rectangular coil within a uniform magnetic field (T = NAIB) The principle of the electric motor (dc and ac types)	04H			- Light sources. - Optical transmission grating with normal incidence	02H
	15		- Biot-Savart law - Ampere's law - Magnetic flux density within a long solenoid, long straight wire, and plane circular coil (B = $\mu_0 nI$, B = $\mu_0 I / 2r$, B = $\mu_0 I / 2\pi r$.	04H	- Multiple slit diffraction. - Reflection and refraction at plane interfaces			02H	

				Force between two current-carrying conductors					
	16			- Force on a moving charge in uniform magnetic field - Measurement of specific charge (e/m_0) - The Hall effect - dia, para- and ferro-magnetic materials - Magnetic Shielding - The Lorentz force	04H			- Laws of refraction. - Refractive index	02H
	17		ELECTRO MAGNETIC INDUCTION	- Faraday's and Lenz's laws of electromagnetic induction. - Induced e.m.f. across a conductor moving with velocity v, through uniform magnetic field. - Simple DC and AC generator principles.	04H		3	- Dispersion. - Total internal reflection and critical angle. - Lenses: principal focus, focal length.	02H
	18			INTEGRATION EXERCISES AND EVALUATION					
	19		ELECTROMAGNETIC INDUCTION + ALTERNATING CURRENT	- Meaning of self-inductance and mutual inductance. - the L-R DC circuit (the growth and decay of current) - Energy stored in an inductor - Coefficient M and L should be understood to be defined as the constants in the equations: $N\phi = LI$ $N\phi_2 = MI_1$ and $N\phi_1 = MI_2$	04H		3.	- Familiarity with practical situations in which a single converging lens produces a magnified or diminished image, diopetre. - Prisms, optical instruments eg compound microscopes and Astronomical telescopes	02H
	20			- Theory of transformers - Knowledge, understanding but not derivation of $\frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$ for ideal transformers	04H			- Conservation of energy for waves in free space from a point source. - Inverse square law	02H

				Sources of power losses for practical transformers and how each is minimized.					
	21			- Root-mean-square values - Impedance and resonance - use of $f_0 = \frac{1}{2\pi\sqrt{LC}}$	04H			- Photoelectric effect. - The photons, Plank constant	02H
	22			- Power in a.c. Circuits. - Rectification of ac signals and Smoothing	04H			- Einstein's photoelectric equation. - Wave-particle duality	02H
	23		The atom, the nucleus of the atom,	The atom, Rutherford's alpha scattering experiment and atomic model.	04H		3.3	- Emission and absorption spectra. - Energy levels. - The electron volt.	02H
	24		HARMONISED EVALUATION						
III	25		Radioactivity	- Nuclear stability and radioactivity - Properties of radiations and applications	04H	Option 2: Communication	3.3 E	- Use of equation: $E = hf = E_2 - E_1$. - The Schrodinger model of the hydrogen atom	02H
	26			- Mass defect in nuclear processes and relation to energy - Nuclear fission and fusion	04H			Representing information: Analogue method, digital method, advantages and disadvantages. Radio waves: Surface or ground wave.	02H
	27	:	Option 3: Electronics	Thermionic emission and the electron gun	04H			- Sky wave, space wave. - Aerials: transmitting, receiving aerials. - Tuning circuit, its resonance curve	02H
	28			- Semiconductors and doping - The p-n junction diode, biasing and applications	04H	Option 2: Communication	3.4	Transmission of information: modulation, demodulation. Amplitude modulation (AM), frequency modulation (FM), advantages and disadvantages of each over the other	02H
	29			- The transistor - Logic gates and amplifiers	04H			- Analogue and digital signals - Analogue to Digital converters - Digital to Analogue converters. Advantages of each over the other	02H
	30		Option 4: Medical Physics	- The physics of vision and defects - Hearing and defects	04H			Main parts. Communication Channels - band width - sidebands - use of satellite for communication Base station and their role	02H

	31			• Biological Measurements for the heart • Imaging in medical diagnosis (Non-ionising methods)	04H			REVISION	02H
	32			• Ionizing technics in imaging for medical diagnosis • Use of optical fibres in medical procedures.	04H			REVISION	02H
	33			REVISION	04H			REVISION	02H
	34			REVISION	04H			REVISION	02H
	35								
	36								

NB: This National Harmonized Progression could be modified in terms of; lesson titles, time allocated for lessons and order of succession of lessons. Such modifications need to be pointed out and discussed with Regional and National Pedagogic Inspectors so that the modifications are communicated to go into effect nationwide as from the same time.